GUI Features and Capabilities

Main Dashboard

- Warning Alerts
    - Sensors inoperative
    - Too high/too low of the variables
    - Instrument uncalibrated
- Overview
    - Sensor readings at the time
    - Most recent photo taken and classification of its stage
- Instrument Auto-Calibration (?)
    - This was suggested by Sir. Ledesma

Monitoring Tab

This tab should focus on providing the details. Sensor readings, humidity, ph level. With each of these variables having their own graphs as a function of time.

- Temperature (DHT11)
    - Since we have 3, we might need to highlight average reading.
- Humidity
- pH Level
- TDS Nutrients

Suggestions: In improving visual aid, we can utilize colors. For temperature, rather than just text, an addition of color gradient to imply low/high temperature would it understandable at first glance. Similarly, humidity could imply a “filling” block depending on humidity level. For pH level, the block could change colors depending on the scale.

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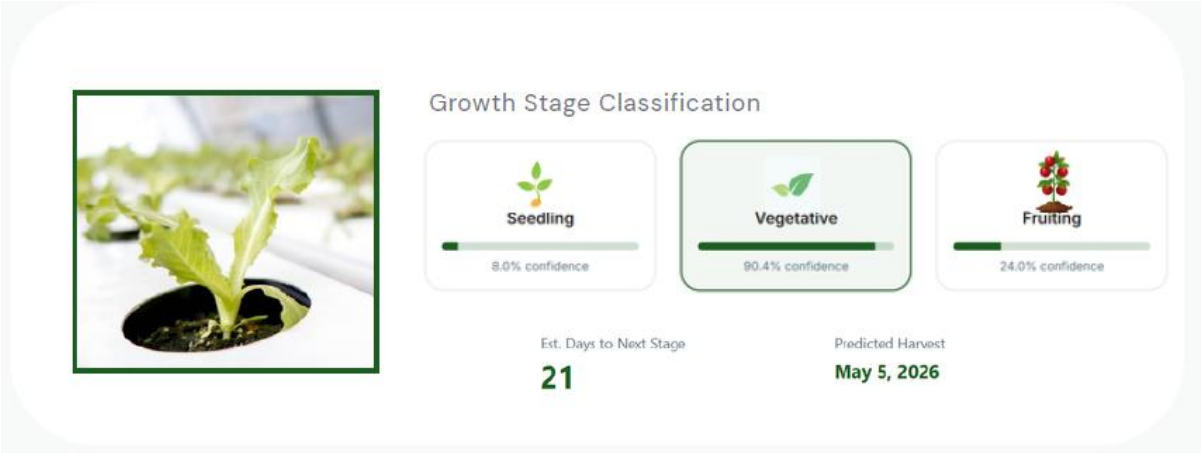
Camera Tab

- Displays most recent photo and stage classification with confidence level.
- 

- Mr. Ledesma suggested that we implement "nutrient" readings as well via photo. That seems improbable at bird's eye view, so my take was to revise the feature having a user-based nutrient updates. Then have our system cross check with our related lits whether they are ideal/sufficient for the stage they are in.
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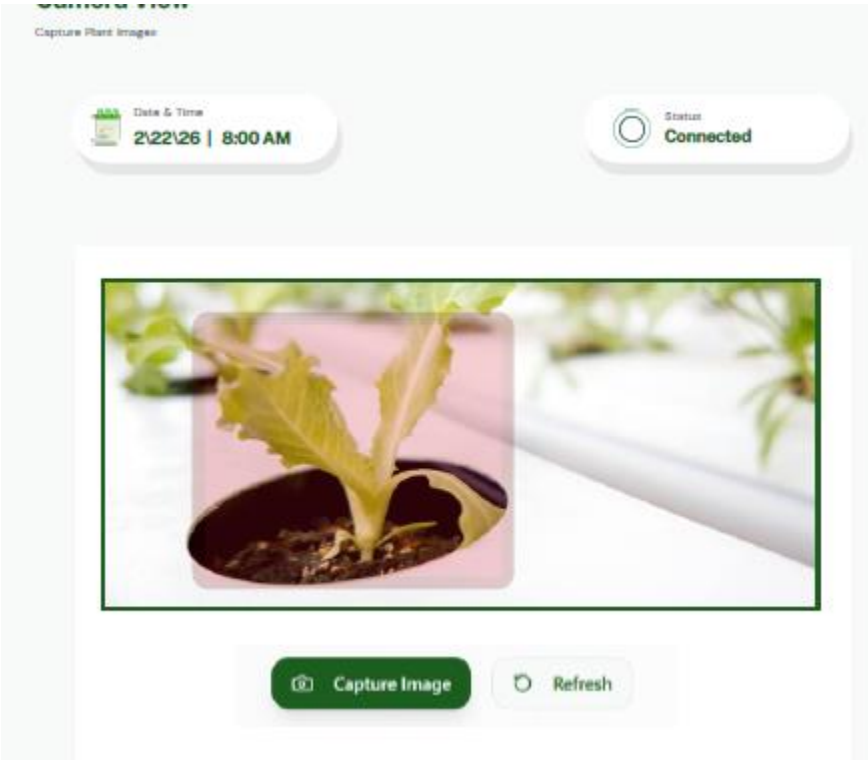
Previously, we had separate tabs for growth analysis and camera view. For simplicity, we can integrate the two into a single tab and call it Camera Analysis.



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- Simply put, instead of just displaying the most recent photo taken, users can capture new photos on the same tab and update real-time.

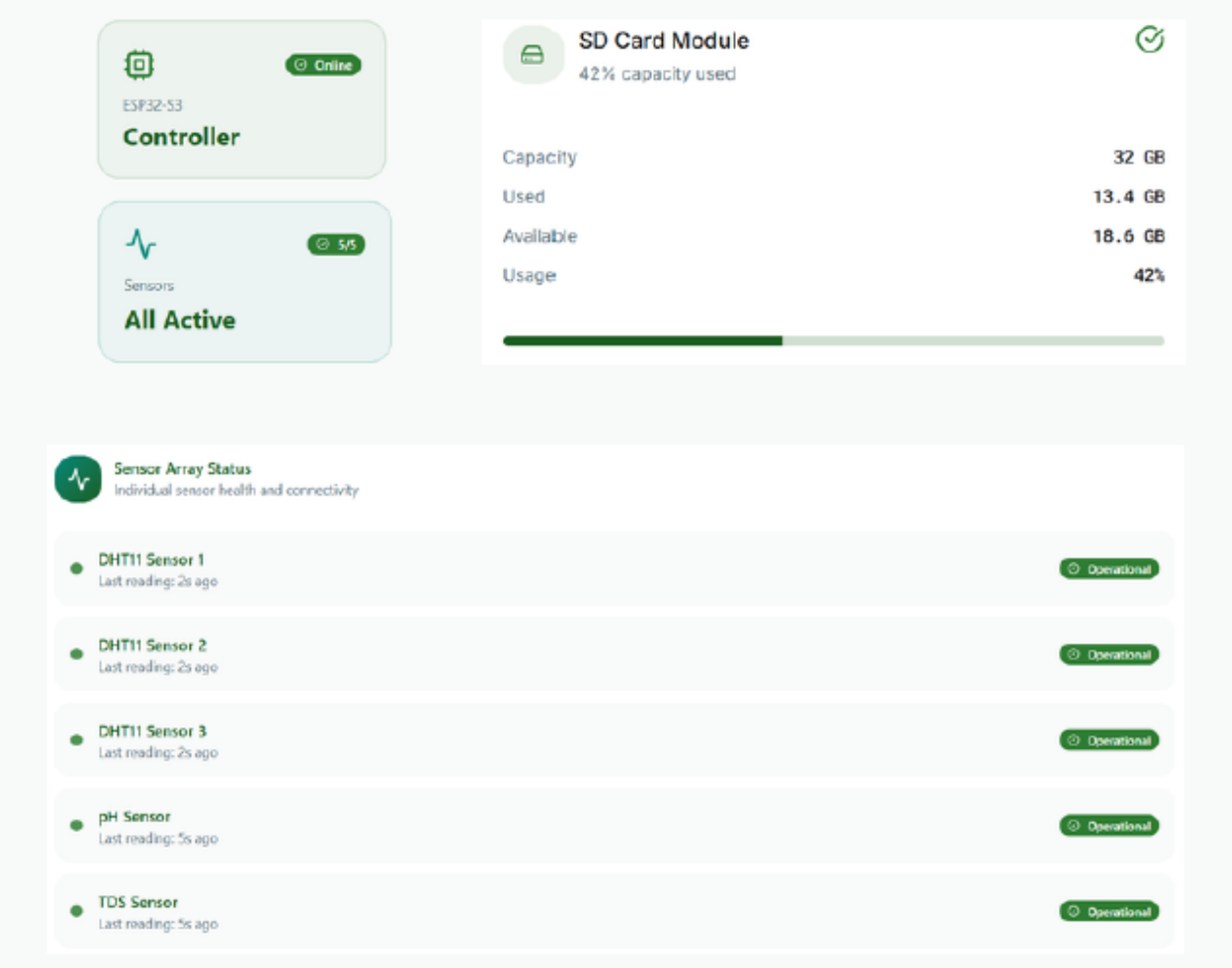
Records & Logs

Displays log history, date and timestamp, variable readings and status.

Timestamp	Temp (°C)	Humidity (%)	pH	TDS (ppm)	Status
2/16/2026, 2:11:53 PM	23.5	61.3	6.96	925	Optimal
2/16/2026, 3:11:53 PM	25.3	55.8	6.04	730	Optimal
2/16/2026, 4:11:53 PM	23.4	61.9	6.16	861	Optimal
2/16/2026, 5:11:53 PM	25.1	55.1	6.35	713	Optimal
2/16/2026, 6:11:53 PM	22.3	58.9	6.31	628	Optimal
2/16/2026, 7:11:53 PM	26	69.7	6.61	954	Optimal

System Health

Displays whether we are receiving data from our sensors. Should also display SD Card storage availability, whether we are connected to the ESP32 module. Block color should change from green to red accordingly.


Online
ESP32-S3
Controller
5/5
Sensors
All Active
SD Card Module
42% capacity used
Capacity 32 GB
Used 13.4 GB
Available 18.6 GB
Usage 42%
Sensor Array Status
Individual sensor health and connectivity
DHT11 Sensor 1
Last reading: 2s ago
Operational
DHT11 Sensor 2
Last reading: 2s ago
Operational
DHT11 Sensor 3
Last reading: 2s ago
Operational
pH Sensor
Last reading: 5s ago
Operational
TDS Sensor
Last reading: 5s ago
Operational